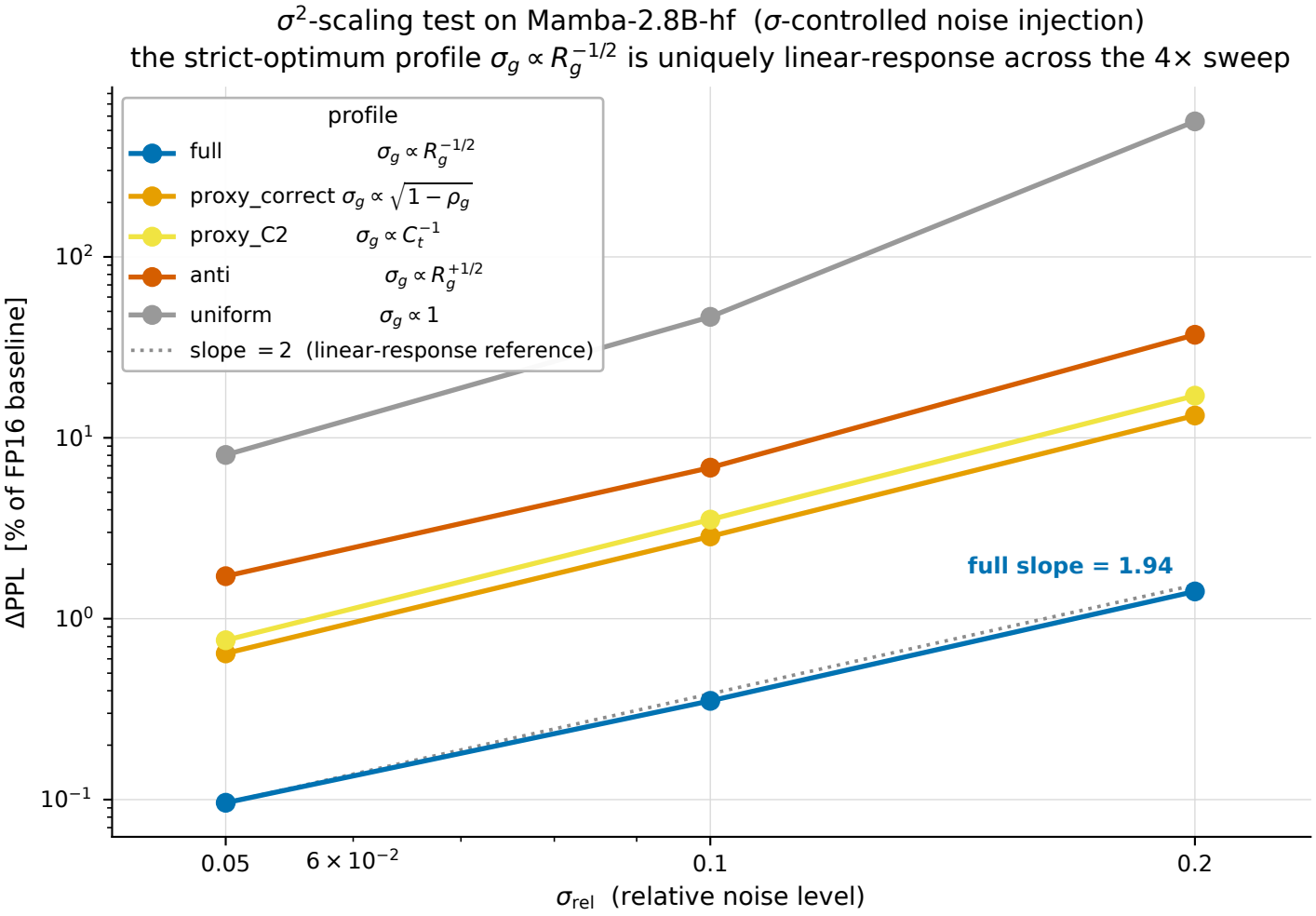
σ^2-scaling test on Mamba-2.8B-hf (σ-controlled noise injection)
the strict-optimum profile $\sigma_g \propto R_g^{-1/2}$ is uniquely linear-response across the 4× sweep
profile
full $\sigma_g \propto R_g^{-1/2}$
proxy_correct $\sigma_g \propto \sqrt{1-\rho_g}$
proxy_C2 $\sigma_g \propto C_t^{-1}$
anti $\sigma_g \propto R_g^{+1/2}$
uniform $\sigma_g \propto 1$
slope = 2 (linear-response reference)
full slope = 1.94
ΔPPL [% of FP16 baseline]
10^2
10^1
10^0
10^{-1}
0.05
6×10^{-2}
0.1
0.2
σ_{rel} (relative noise level)